POWERED BY LOGIC GATES AND SENSORS

THE GATE PROTOCOL

RESONANCE 2026

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(HARDWARE HACKATHON)

OVERVIEW:

- Day 1 - Hardware / Embedded systems quiz, Day 2 - Work day
- Work day timing: 8 Hours (9 AM to 5 PM)
- Team size: 2-3 (No more than 3 members in a group)
- Teams get 1 hour to use AI chatbots, AI IDEs, or any AI-related tools/plugins to help with coding (9 AM to 10 AM), after which volunteers will ensure no AI is used. Internet access remains available throughout.
- All teams receive one project idea to work around, revealed on the spot, the team that best materialises the idea wins.
- Participants are required to bring their own laptops with Arduino IDE installed.

PRIZE POOL : ₹15,000(1st: ₹7,000 |2nd: ₹5,000 |Higher Secondary:₹3,000)
ENTRY : ₹750

COMPONENTS PROVIDED BY COORDINATORS:

- Breadboard (for testing concepts)
- Perf board (for soldering)
- Jumper cables (Bread board)/Single core wires (Soldering)
- Electronic items: Microcontrollers, Sensors, Actuators, Push buttons, Switches (Revealed and given on work day)
- Housing materials: Thermacol, Plastic boards, etc. (if required)
- Cable for microcontroller
- Solder iron

COMPONENTS NOT PROVIDED BY COORDINATORS:

- Glue gun
- Solder and De-soldering copper wick, flux
- Power supply / Battery to power microcontroller

RULES & REGULATIONS:

- No use of AI after the stipulated time (after 10 AM).

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- Participants will be given a bonus problem midway (2 PM), which they can choose to implement or not. Correct and working implementation earns bonus points.
- Judging will be based on functionality, completeness, finishing, and bonus problem implementation, in that order.

POINT WEIGHTAGE:

Priority: FUNCTIONALITY > COMPLETENESS > FINISHING > BONUS

CRITERION	DESCRIPTION
FUNCTIONALITY	A working prototype on a breadboard scores more than a soldered but non-functional project.
COMPLETENESS	Are all provided components used? Does the project perform all intended functions from the problem statement?
FINISHING	Visual quality, soldering quality, form factor, neatness, and adherence to defined constraints.
BONUS PROBLEM	An optional addition that boosts points if implemented correctly. Choosing to attempt and failing the bonus problem results in a point deduction equal to the bonus value.

NOTE: The detailed scoring breakdown will be provided to all teams at the start of the event.
Lunch will be provided on Day 2

CO-ORDINATORS:

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